3) What is the relevance of examining GSDP growth rates to understand regional disparity?

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24.4 AGRICULTURAL DEVELOPMENT AND REGIONAL DISPARITY

As discussed, the contribution of agricultural sector to the total GDP has been gradually decreasing over time. Agriculture's contribution reduced from 51.9 to 16.5 per cent during the period 1950-51 to 2019-20. Even over a shorter period of time there has been a significant decline in the share of agriculture in the Gross State Domestic Product (GSDP).

Table 24.2: Share of Agriculture to GSDP and Total Workforce in India and States

State	Share of Agl in GSDP (% at 2011-12 price)		Share of Agriculture in Total Workforce (%)	
	2011-12	2018-19	2011-12	2018-19
Andhra Pradesh	13.7	11.2	52.4	43.9
Assam	14.4	11.5	54.3	37.2
Bihar	17.2	10.8	61.6	47.6
Chhattisgarh	11.4	9.9	71.8	60.8
Delhi	0.2	0.0	0.1	0.3
Gujarat	12.9	6.4	46.9	39.4
Haryana	13.7	8.1	40.9	24.3
Himachal Pradesh	9.3	5.8	58.4	56.3
Jammu & Kashmir	10.2	7.1	40.3	35.2
Jharkhand	8.9	5.9	49.4	40.3
Karnataka	8.8	5.2	48.5	38.9
Kerala	8.0	4.1	20.4	16.8
Madhya Pradesh	23.2	21.5	57.7	55.7
Maharashtra	8.0	5.1	49.1	41.8
Odisha	11.1	7.0	54.8	42.8
Punjab	18.8	13.3	35.8	24.0
Rajasthan	16.9	11.5	49.9	51.9
Tamil Nadu	7.2	3.9	33.5	25.3

Telangana	9.0	4.9	-	41.8
Uttar Pradesh	17.1	13.3	51.9	49.7
Uttarakhand	6.6	3.7	46.6	31.8
West Bengal	13.9	12.1	36.8	32.6
India	11.5	7.8	47.8	40.9

Source: Calculated from MOSPI and NSS PLFS, 2018-19

The contribution of agricultural sector to total GSDP in 2011-12 was 11.5 per cent which reduced to 7.8 per cent in 2018-19. The state level picture also shows the same declining trend in the contribution of agricultural sector. In the year 2018-19, the contribution of agricultural sector to state's GSDP was high in Madhya Pradesh (21.5 per cent) followed by the state Punjab (13.3 per cent). The change in the contribution of agriculture in GSDP over the period 2011 to 2018 was highest in Bihar and Gujarat (about 6 percentage point reduction) whereas Delhi, Madhya Pradesh and West Bengal registered the lowest percentage point change (see Table 24.2).

On the other hand, the percentage share of agriculture in the total workforce has only reduced from 48 per cent in 2011 to 41 per cent in 2018. A large disparity is visible among the states in terms of contribution of agriculture in total workforce. The states showing the highest proportion of agricultural workforce to total workforce are Chhattisgarh (71.8 per cent), Bihar (61.6 per cent), Himachal Pradesh (58.4 per cent) and among the lowest are Kerala (20.4 per cent), Tamil Nadu (33.5 per cent) and Punjab (35.8 per cent).

The percentage point reduction in the contribution of agricultural workforce in the total workforce over the period 2011 to 2018 was highest in Assam (17.1 percentage point), Haryana (16.6 percentage points) and Uttarakhand (14.8 percentage points) whereas in the states Uttar Pradesh (2 percentage points), Himachal Pradesh (2.1 percentage points), Madhya Pradesh (2 percentage points) the percentage point reduction was the lowest. The state Rajasthan shows an increase of percentage of workforce in agriculture from 2011 to 2018.

Figure 24.2 explains the growth rate of Domestic product from agriculture from 2004-05 to 2018-19. The growth rate of agriculture in India from 2011-12 to 2018-19 is 8.2 per cent which increased from 4.4 per cent during 2004-05 to 2011-12. Among the states the highest growth rate during 2011-12 to 2018-19 is marked by Madhya Pradesh (15.3 per cent), Maharashtra (10.6per cent), followed by West Bengal (10.3 per cent). During 2004-05 to 2011-12 Jharkhand registered the highest growth rate of 9.5 per cent followed by Chhattisgarh 7.9 percent. Kerala registered the negative growth of 1.1 per cent during 2004-2011.

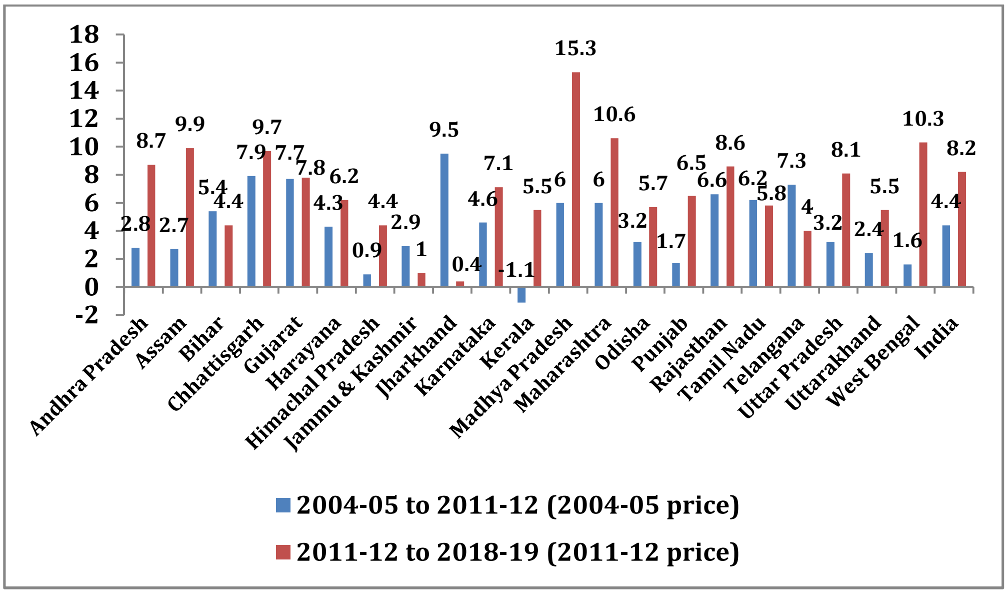

Fig. 24.2:Growth Rate of Domestic Product from Agriculture, 2004-05 to 2018-19

Source: MOSPI

A sectoral disaggregation of the workforce in rural areas shows that the share of agricultural sector to the total workforce has declined but at a much slower pace than the decline in the share of agriculture in GSDP. As expected, there has been a decline in the share of workforce in agriculture from 81.4 per cent to 58.2 per cent between 1983 to 2018-19. During this period, the dependence of workforce declined by 29 percentage point. Considering the state-wise contribution of farm sector to total employment,in 2018-19, among the major states Chhattisgarh and Madhya Pradesh contributed the highest proportion of farm employment, whereas in the states like Kerala and West Bengal, the share of farm sector in employment was the lowest.

Figure 24.3 explained the proportion of agricultural labour and non-agricultural labour over the period of 1983 to 2018-19. The figure clearly shows that the proportion of non-agricultural labour increased from 18.6 per cent in 1983 to around 42 per cent in 2018-19. On the other hand, the proportion of worker in agricultural sector reduced from 81.4 per cent in 1983 to around 58 per cent in 2018-19. This clearly indicate that the increase in proportion of non-agricultural workforce in 2018-19 was more than two times compared to what it was in 1983. On the other hand, the reduction of agricultural workforce was only one third in 2018-19 compared to what it was in the year 1983.

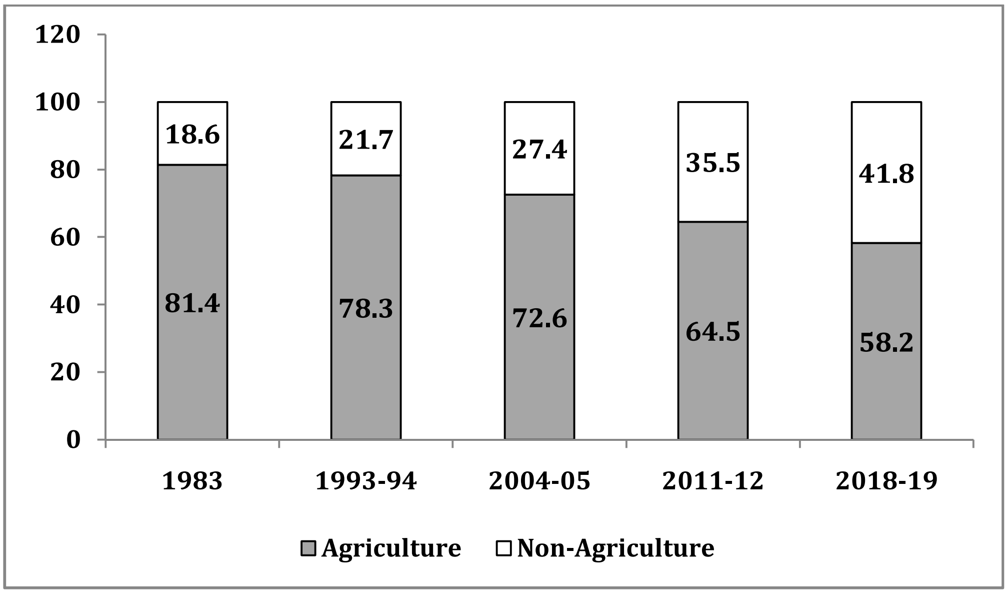

Fig. 24.3: Percentage Distribution of Workers

Source: Calculated from different round of NSS Employment unemployment rounds

The labour productivity in agriculture has been growing very slowly. The slow growth is largely due to the following factors: (i) the impact of green revolution was limited to a very few states and (ii) the process of diversification of labour from agriculture to non-agriculture has barely started.

In short, agricultural performance has varied among different states, accounting partially though significantly, for prevalent wide disparities in income among different states.

24.5 INDUSTRIAL DEVELOPMENT AND REGIONAL DISPARITY

Like other developing countries, industrial concentration is observed in some pockets of India. Keeping this in view, Government of India has adopted a plethora of measures to achieve a balanced regional development. The policies are guided by industrialisation mixed with highly regulated policies. Also, many industries reserved for public sector. After opening up of the economy with minimum role of the state in industrialisation and industrial growth, it has been argued that the industries have concentrated in the economically advanced states due to their comparative advantages in social and economic infrastructure. This argument has been supported by several country level studies.

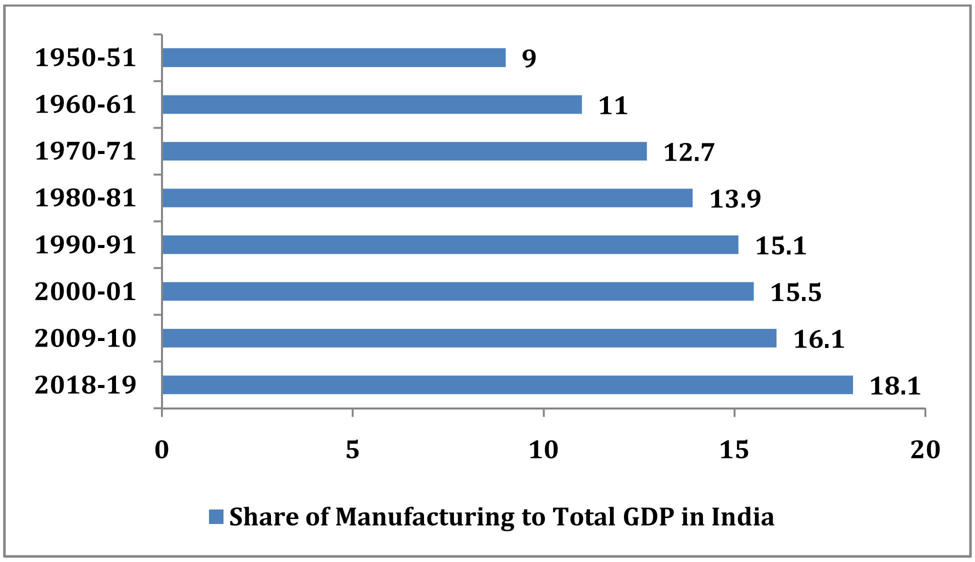

Fig. 24.4: Share of Manufacturing to total GDP in India

Source: RBI; downloaded from https://www.rbi.org.in

The share of manufacturing in total GSDP in 2018-19 in major states ranges between 5.1 per cent in Delhi to 37.2 per cent in Uttarakhand(refer Table 1 in Appendix 24.1). The industrially developed states like Gujarat and Maharashtra also showed a higher contribution of manufacturing in total GSDP of the state. In the year 2018-19, the contribution of manufacturing to GSDP in 10 states reduced as compared to the year 2011-12 whereas in 12 states the share of manufacturing increased. The states registering highest reduction in the share of manufacturing to total GSDP were Telangana (5.0 percentage points) and Andhra Pradesh (3.1 percentage points). The states showing the highest increase were Gujarat (7.1 percentage points) and Himachal Pradesh (5.2 percentage points).

The growth rate of Industry has been 8.5 per cent between 2011-12 to 2018-19. The highest growth rate is of Gujarat (13.8 per cent) followed by Assam (12.9 per cent) whereas the states registering the lowest growth rate are Rajasthan (2.6 per cent) and Telangana (2.8 per cent).

Similar conclusions can be drawn when we analyse the trend in share of service sector in GSDP of different states.

The contribution of service sector increased from 44.8 per cent to 46.8 per cent between 2011-12 to 2018-19 (refer Table 2 in Appendix 24.1). This means there is a 2 percentage point increase in the said period. In the year 2018-19, Delhi (73.6 per cent) and Bihar (57.7 per cent) shows the highest share in service sector and Gujarat (30.8 per cent), Chhattisgarh (34.5 per cent) and Uttarakhand registered the lowest share in service sector.

The percentage point change among states between 2011-12 to 2018-19 shows that Telangana and Jharkhand recorded the highest percentage point change of 7.9 and 4.1 respectively, whereas Himachal Pradesh, Arunachal Pradesh and Jammu and Kashmir show a low change in share of service sector (1.00, 1.00, and 1.4 respectively). Again Odisha, Delhi and Madhya Pradesh show no change in the contribution of service sector in GSDP in between 2011-12 to 2018-19.

The growth rate of service sector is 7.9 per cent between 2011-12 to 2018-19. The highest growth rate is marked by Telangana (10.0 per cent) followed by Karnataka (10.0 per cent) whereas the states registering the lowest growth rate include Assam (5.3 per cent) and Jammu and Kashmir (5.8 per cent).

24.6 INFRASTRUCTURAL DEVELOPMENT AND REGIONAL DISPARITY

The importance of infrastructure in economic development, trade, employment and in reducing disparity within the country/region cannot be over emphasised. As learnt extensively in Unit 4 availability of adequate infrastructure facilities, especially the physical infrastructure is the pre-condition for sustainable economic and social development. Non-availability or inadequate availability of infrastructure poses a serious threat to growth. We examine the prevalent situation in regard to available infrastructure in different states of the country.

24.6.1 Social Infrastructure

The social infrastructure broadly includes health, education. The indicators used to access the availability of education infrastructure in India are number of elementary schools (primary schools and upper primary schools) across states. These indicators are converted on per 10,000 population basis to facilitate comparison among states.

Education Infrastructure

It is well recognised that the literacy of any region or area has a positive relation to the overall development. It enables people to access new opportunities to participate in society in different ways. The census data shows that the literacy rate increased from 18.3 per cent in 1951 to 74.04 in 2011.

The education infrastructure plays an important role in the literacy rate of a particular region. The supply side factor, like the availability of school, good condition building, adequate number of teachers, facility like toilet, drinking water are the major determinants of the quality of education within a region. Figure 24.5 shows the number of elementary schools (primary and secondary) per lakh population in 2016-17. The number of elementary schools per lakh population in India is 114. Himachal Pradesh (254), Jammu and Kashmir (223) and Uttarakhand (220) registered a high number of Elementary school per lakh population. On the other hand Delhi (31), Kerala (48), Gujarat (68) registered a low number of elementary schools per lakh population.

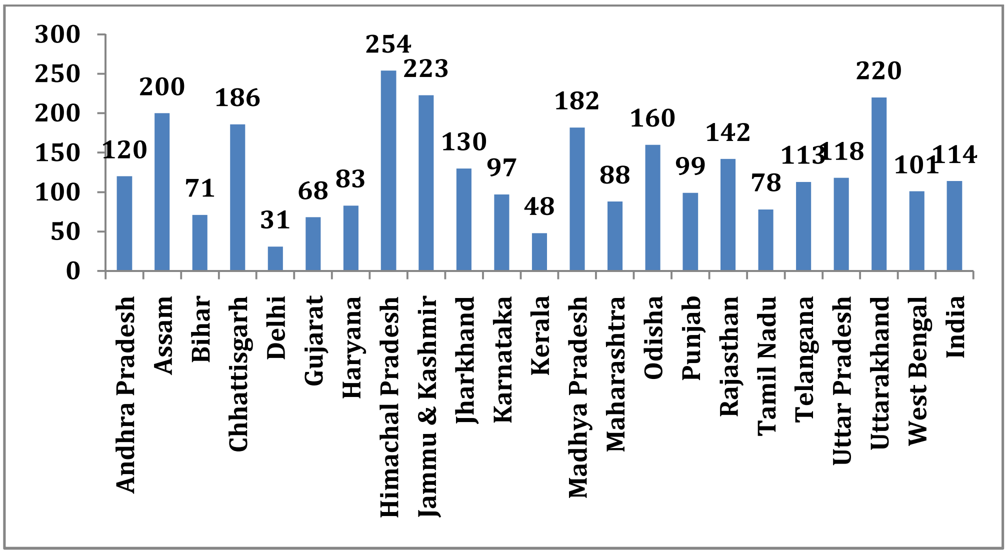

Fig. 24.5: Number of Primary and Upper Primary Schools Per Lakh Population, 2016-17

Note: Total number of school is pertaining to all school up to upper primary level both government and private and for the year 2016-17 and the population for the year 2016 is taken from Census projected population.

Source: DISE www.**dise**.in and Census of India 2011

The regional disparity in education can also be judged from some other characteristics like percentage of single class room school, percentage of single teacher school. In India 4.3 per cent elementary schools have only a single classroom (Refer Table 3 in Appendix 24.1). Among the states Assam (18.5per cent), followed by Andhra Pradesh (13.0 per cent) registered a high proportion of single room schools whereas Kerala, Delhi (0.1 per cent), Uttar Pradesh (0.7 per cent), Tamil Nadu (0 per cent) show a low proportion of school with single class room. The other important indicator is the single teacher school. In India 7.2 per cent of schools have only a single teacher. Among the states, Andhra Pradesh (14.1 per cent), Jharkhand (17.1 per cent), and Rajasthan (12 per cent) registered a high single teacher school. Delhi (0.1 per cent) and Gujarat (2.0 per cent) have low proportion of single teacher schools in the year 2016-17.

Some of the basic facilities in schools also influence the educational development. These facilities include toilet facility, drinking water, girl's toilet facility, condition of school building. In India 96.5 per cent of schools are having girl's toilet facility. Also, in terms of drinking water facility a very low variation was found among states. In terms of electricity facility about 60 per cent schools have electricity connectivity. States with a high proportion of schools with electricity connectivity are Delhi, Gujarat and Haryana (registering 94 to 99 per cent of schools) whereas states with low proportion of school with electricity connectivity were Assam, Madhya Pradesh, Jammu and Kashmir and Jharkhand (ranging from 20 to 35 per cent of schools).

One of the important indicators of education is the pupil-teacher ratio (PTR). The higher the PTR, the lower the quality of education imparted to children. PTR in India as a whole was 23 whereas states registering highest PTR were

Bihar (45), Jharkhand (32) whereas states showing low PTR were Himachal Pradesh, Jammu and Kashmir and Punjab (below 16).

To sum up, the above brief review helps to bring out that (i) the existing social infrastructure in India is weak and inadequate, and (ii) wide disparities prevail between different states.

Health Infrastructure

The availability of health infrastructure shows the status of health sector in the country. The total sub-centres per million population in rural India was 179. The states showing higher number of Sub-centres were Kerala (458), Jammu and Kashmir (325) whereas states showing lowest Sub-centre per million population were Bihar (95), Jharkhand (138) (Refer Table 4 in Appendix 24.1).

The primary health centre (PHC) per million population in India is 28. The state showing the highest number of PHCs were Himachal Pradesh (89) followed by Kerala (72) whereas states showing least number were Jharkhand (11) and Bihar (18). The availability of doctors in PHCs per million population was 34 in India and a large variation was found among states. States having the highest number of doctors was Kerala (130) followed by Jammu and Kashmir (99) and states having lowest were Jharkhand (12) and West Bengal (13). The Auxiliary Nurse Midwife (ANM) particularly in rural areas plays important role in facilitating health services as well as imparting general health awareness. The number of ANM per million population in India was 266 and states having the highest number of ANM were Kerala (638) followed by Andhra Pradesh (343). States showing the lowest number of ANM per million population were Uttar Pradesh (169) followed by Madhya Pradesh (201).

24.6.2 Physical Infrastructure

The physical infrastructure includes transport, communication, electricity etc. As observed earlier in Unit-4, India suffers from inadequate availability of physical infrastructure, as measured by any accepted indicators. Not only is infrastructure inadequate and weak, it varies from state to state, and even within each state from district to district.

The per capita consumption of electricity is one of the important indicators of development. In 2018, the per capita electricity consumption for India was 1181 kWh, which is quite low when compared with many countries like Canada kwh 15438 kWh, USA 13098 kWh, Saudi Arabia 10239 kWh. The per capita electricity consumption by states, shows a huge disparity between the various states. The developed states like Gujarat (2378 kWh), Haryana (2082 kWh), Punjab (2046 kWh) show the highest per capita consumption of electricity. The states like Bihar (311 kWh), Assam (341 kWh) show the lowest consumption. The ratio of consumption of electricity between Gujarat and Bihar was more than 7 times.

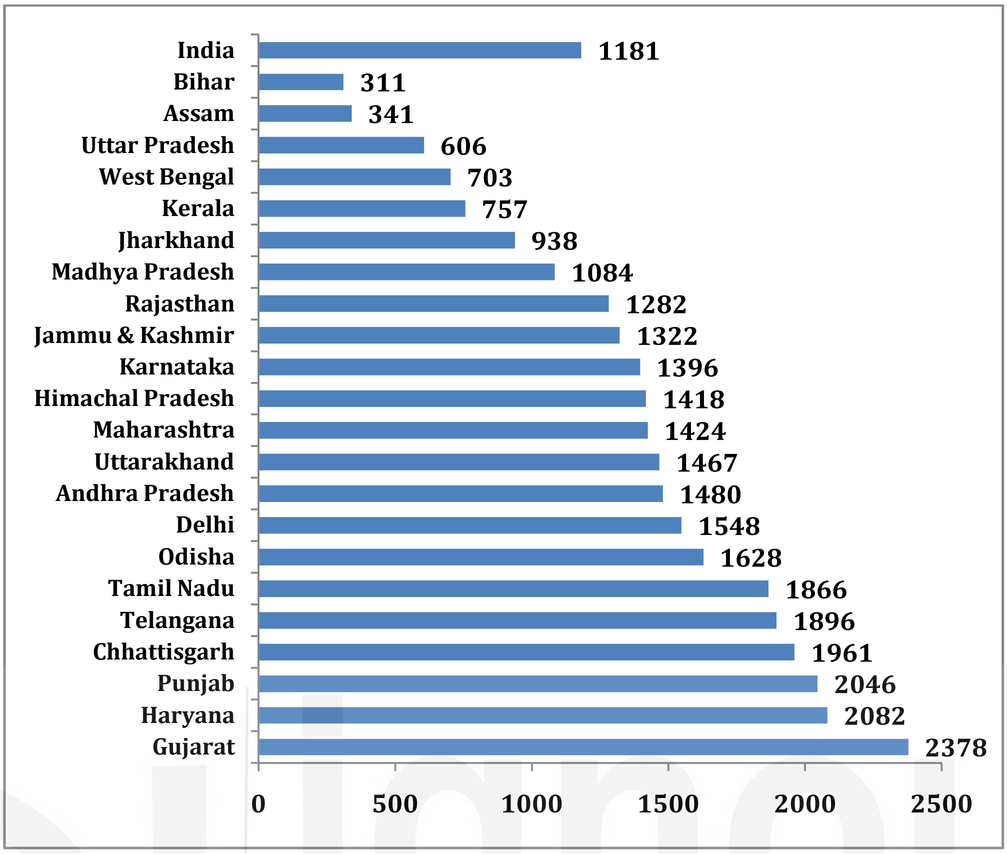

Fig. 24.6: Per Capita Electricity Consumption, 2018-19

Source: Electricity Authority of India

Check Your Progress 2

1) Name the states where agriculture sector's contribution in GSDP have declined since 1980-81.

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2) What are implications of relatively lower decline in the share of agricultural sector's workforce to total workforce of a state relative to the decline in the share of agriculture sector SDP in that states GSDP?

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3) State whether regional disparities in industrial development have declined over time.

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4) State the indicators of educational and health development.

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24.7 HUMAN DEVELOPMENT AND REGIONAL DISPARITY

Talking of the disparity in Human Development Index (HDI), Kerala (0.779) maintained the first place in 2018 followed by Delhi (0.746) (refer Table 5 in Appendix 24.1). On the other hand, the states showing the lowest position in HDI were Bihar (0.576) and Uttar Pradesh (0.596).

Over the period of analysis, the states Delhi, Kerala, Punjab were the states which were maintaining their position in the upper strata in the HDI index whereas the states Bihar and Uttar Pradesh were the states which remained in the lower strata of HDI index (UNDP HDR different years[1]).

24.7.1 Causes of Regional Disparities

In India – just and in majority of the developing world – regional disparities date back primarily to the colonial rule. It was at that time that the areas lying farther away from the seacoast began to fall behind those close to it, though the differences in the state of development were also influenced by other factors which cannot be attributed to the colonial period alone. The important factors can be summarised as follows:

1) Disparities in per capita income can be explained in terms of the economic sector thesis of Colin Clark; it maintains that the levels of income are higher in those regions where a larger proportion of working population is engaged in manufacturing and tertiary sectors. Per capita

[1] Downloaded from https://globaldatalab.org/shdi/shdi/IND/?levels=1%2B4&interpolation=0&extrapolation=0&nearest_real=0